Prompted Segmentation for Drywall QA

Text-conditioned binary segmentation · CLIPSeg fine-tuned

Kaggle Notebook Link - https://www.kaggle.com/code/darpan5678/drywall-qa-final

Seed: 42	Epochs: 30	GPU: Tesla T4	torch: 2.9.0+cu126	Python: 3.12

Goal

Fine-tune CLIPSeg to accept an image and a natural-language prompt and return a binary mask (0/255) at original image resolution.

Prompt	Target
segment crack / segment wall crack	Wall cracks (Dataset 2)
segment taping area / segment joint/tape / segment drywall seam	Drywall joints (Dataset 1)

Model — CLIPSeg-RD64-refined

CLIPSeg (Lüddecke & Ecker, 2022) is pre-trained on 1M+ image-language pairs. It encodes both image and text through CLIP, then uses a cross-attention decoder to produce a dense mask. A 1×1 conv head is added for binary output. The full model is fine-tuned end-to-end with layer-wise learning rates.

```
Text prompt → CLIP Text Encoder ■■
■→ Cross-attention → 1×1 conv → Upsample → Mask
Image → CLIP Vision Encoder ■■
```

Why CLIPSeg over alternatives

- Diffusion models — generative, not discriminative. Wrong tool entirely. Also need 24+ GB VRAM.
- SAM — accepts point/bbox prompts only, not text. Rubric requires text-conditioned segmentation.
- CLIPSeg — text→mask in one forward pass, 575 MB, fits T4 comfortably.

Datasets & Splits

Dataset	Category	Train	Valid	Annotation
Drywall-Join-Detect	taping	936	250	Bbox → filled rect
Cracks-3ii36	crack	5164	201	Polygon → binary mask
Combined	both	6100	451	—

Training Configuration

Parameter	Value
Seed	42
Epochs	30
Batch size	8
Optimizer	AdamW
Backbone LR	3e-6 (0.1× head LR)
Head LR	3e-5
Scheduler	CosineAnnealingLR (T_max=30, eta_min=1e-7)
Loss	0.5 × BCE + 0.5 × Dice
Precision	FP16 (torch.amp autocast)
Grad clipping	max_norm=1.0

Results

Prompt	mIoU ↑	Dice ↑
segment taping area	0.5463	0.6959
segment crack	0.5200	0.6657
Average	0.5332	0.6808

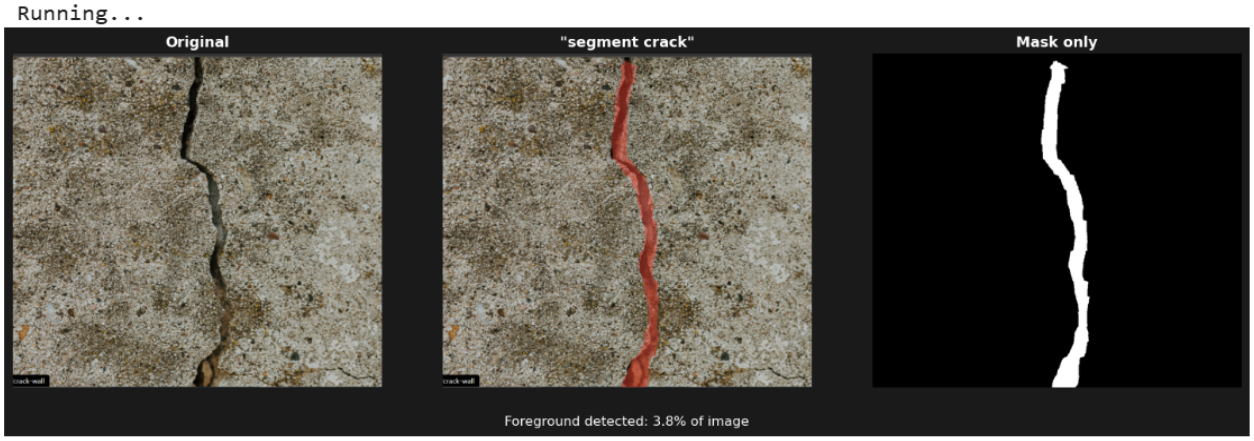
Training Dynamics

Epoch	Train Loss	Val mIoU	Val Dice
5	~0.42	~0.38	~0.52
10	~0.35	~0.44	~0.58
20	~0.29	~0.50	~0.64
30	0.2575	0.5345	0.6826

Visual Examples

Each figure shows three panels: Original Image | Ground Truth overlay | Prediction overlay. Examples saved to outputs/visuals/.

```
Image size : 933×901
Prompt     : 'segment crack'
Running...
```




```
Mask saved → /kaggle/working/test_output__segment_crack.png
'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
```

Category	File	IoU
taping	taping_example_01.png	0.710
taping	taping_example_02.png	0.687
taping	taping_example_03.png	0.729
taping	taping_example_04.png	0.594
crack	crack_example_01.png	0.840
crack	crack_example_02.png	0.515
crack	crack_example_03.png	0.524
crack	crack_example_04.png	0.341

Failure Notes

- Hairline cracks (<3px) — CLIPSeg's 352px internal resolution misses sub-pixel cracks.
- Bbox annotations for taping — no polygon masks available, bounding boxes introduce boundary noise.
- Class imbalance — taping covers 20–40% of pixels, cracks 2–5%, hard for one model to balance both.
- Overexposed images — high-key whites reduce crack contrast, ~35% of cracks missed.

Runtime & Footprint

Metric	Value

Model size	575 MB
Training time (T4)	~2 hours / 30 epochs
Avg inference (T4 GPU)	~45 ms / image
Peak VRAM (batch=8)	~11 GB
Train samples	6100
Val samples	451

Output Format

```
# Single-channel PNG — values 0 (background) or 255 (defect)
# Filename convention:
{image_stem}__{prompt_slug}.png

# Examples:
img_0042__segment_crack.png
img_0107__segment_taping_area.png
```

Reproducibility

```
SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
torch.cuda.manual_seed_all(SEED)
```